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Research paper

Comparison of demand response strategies using active and passive thermal energy storage in a food processing plant

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ABSTRACT

Demand side management holds potential for improving energy efficiency and cutting energy consumption within the food industry. This research introduces a demand response approach tailored for an industrial food processing facility, utilizing a chilled water buffer as active thermal energy storage and the plant building as passive thermal energy storage. The plant building and production process are modeled using transient thermal energy balances and the demand side management problem is formulated as a linear program. Model predictive control is employed to manage uncertainties in the optimization process. A simulated case study of an Austrian food processing plant shows reductions in electrical power consumption by up to 18%, electricity costs by up to 24%, and peak load by up to 36% in three distinct optimization scenarios. Simple prediction approaches via averaging historical data already lead to nearly optimal results concerning energy consumption and cost reduction. Highly accurate predictions are necessary for peak load reduction, as considering the simple prediction method only roughly a third of the potential reductions are achieved.

1. Introduction

Food production is an energy-intensive industrial subsector (Gerres et al., 2019), in which energy costs account for about 20%–50% of the food production costs (Clairand et al., 2020). One promising method to decrease energy costs, decrease emissions, or increase energy efficiency is demand side management (DSM) (Palensky and Dietrich, 2011), which can be categorized in demand response (DR) and energy efficiency (EE) (Zhang and Grossmann, 2015). While DR summarizes all methods of load profile adjustments (e.g. peak shaving, valley filling, or load shifting), EE consists of permanent measures like retrofitting and process design improvements (Gelazanskas and Gamage, 2014).

In the following, existing literature is reviewed, first on EE and afterward on DR in industrial applications, particularly focusing on the food industry. Finally, the research gap is identified.

Morais et al. (2022) investigated the EE and energy consumption of 60 different food processing plants in Portugal, and Lima et al. (2018) investigated the EE and energy mix of 37 dairy cheese-making facilities. Both conclude, that food processing plants offer a high potential for EE improvements. The EE can be improved by integrating renewable energies with solar energy (Nacer et al., 2016a,b; Maammeur et al., 2017) or wind energy (Houston et al., 2014). Also process design improvements and retrofits such as heat recovery (Philipp et al., 2018), improving the thermal insulation (Briceño-León et al., 2021), or installing a heat pump (Becker et al., 2012) pose viable options to improve EE. The EE measures mentioned are design optimizations and permanent measures with high investment costs.

In contrast, in DR the operation is optimized. Operational optimization often uses already available flexibility thereby resulting in lower investment costs. Siddiquee et al. (2021) show the potential of DR for load shifting in industrial applications. In the food industry, this potential stems from the energy necessary for heating and cooling the food product (Morais et al., 2022). This energy demand can either be shifted to times with higher efficiency or lower electricity prices. Typically, load shifting is done via energy storage or process rescheduling. Rescheduling industrial processes is often hard as they rely on various factors, such as production deadlines, working schedules, logistics, or warehousing. In contrast, energy storage systems provide direct DR

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Nomenclature

Indices

i	Discrete time point index
n	Production process index
t	Time (s)

Parameters

α	Absorptance (–)
$\dot{Q}_{\mathrm{S,F}}$	Steam boiler to food product heat transfer rate (W)
$\dot{Q}_{\mathrm{S,P}}$	Steam boiler waste heat transfer rate (W)
$\dot{Q}_{\mathrm{S}}$	Steam boiler heat transfer rate (W)
$\dot{Q}_{\mathrm{dem}}$	Demand heat transfer rate (W)
$\dot{Q}_{\mathrm{Sun}}$	Solar radiation (W)
η_{F}	Efficiency of the heating process (–)
η_{G}	Efficiency of the steam boiler (–)
π	Price function (€/J)
ρ	Density (kg/m^3)
ε_1	Energy efficiency ratio of chiller 1 (–)
ε_2	Energy efficiency ratio of chiller 2 (–)
C_{B}	Thermal capacity of the building envelope (J/K)
C_{C}	Thermal capacity of the cooling hall (J/K)
c_{E}	Specific thermal capacity of the cooling medium (J/(kg K))
c_{F}	Specific thermal capacity of the food product (J/(kg K))
C_{P}	Thermal capacity of the production hall (J/K)
$C_{\mathrm{F},n}$	Thermal capacity of the food product at process step n (J/K)
k_{B2P}	Overall heat transfer characteristics from building envelope to production hall (W/K)
$k_{\mathrm{B2\infty}}$	Overall heat transfer characteristics from building envelope to ambient (W/K)
$k_{\mathrm{F},n}$	Overall heat transfer characteristics from food product at process step n (W/K)
k_{P2C}	Overall heat transfer characteristics from production to cooling hall (W/K)
$K_{\mathrm{p,1}}$	Proportional factor of controller 1 (W/K)
$K_{\mathrm{p,2}}$	Proportional factor of controller 2 (W/K)
$m_{\mathrm{F},n}$	Food product mass (kg)
N_{F}	Number of process steps (–)
N_{T}	Number of time steps (–)
P_{G}	Gas power (W)
T_{∞}	Ambient temperature (K)
T_{E}	Cooling medium temperature (K)
V	Volume (m^3)

Variables

$\dot{Q}_{\mathrm{Ch,1}}$	Chiller 1 heat transfer rate (W)
$\dot{Q}_{\mathrm{Ch,2}}$	Chiller 2 heat transfer rate (W)
$\dot{Q}_{\mathrm{B2P}}$	Heat transfer rate from building envelope to production hall (W)
$\dot{Q}_{\mathrm{B2\infty}}$	Heat transfer rate from building envelope to ambient (W)
$\dot{Q}_{\mathrm{F},n}$	Heat transfer rate from food product at process step n (W)
$\dot{Q}_{\mathrm{P2C}}$	Heat transfer rate from production hall to cooling hall (W)
E	Energy (J)
$P_{\mathrm{El,1}}$	Electrical power of chiller 1 (W)
$P_{\mathrm{El,2}}$	Electrical power of chiller 2 (W)
SOC	State of charge (–)
T_{B}	Building envelope temperature (K)
T_{C}	Cooling hall temperature (K)
$T_{\mathrm{F},n}$	Food product temperature at process step n (K)
T_{P}	Production hall temperature (K)

potential and are easier to use. Arteconi et al. (2012) review and show the potential of TESs (thermal energy storages) for DSM. TESs can be categorized into active and passive. In an active TES, sensible or latent heat is stored in fluids (e.g. chilled water buffer, ice storage). For a passive TES, solid structures with a high thermal capacity, such as thermal building mass or even the industrial product's mass are used. The main benefit of passive TESs is that the thermal capacity is easily used and available, and, therefore, lower investment costs are needed. The benefit of active TESs is that the models needed for implementation are derived and parameterized much easier (usually an energy balance) compared to the models needed for using a passive TES, which requires detailed information regarding system dynamics. The system dynamics are usually modeled via transient thermal energy balances resulting in a coupled system of differential equations.

The following paragraphs provide an overview of the DR literature in various sectors. Table 1 shows an overview of the conducted literature review. The investigated DR literature can be categorized into energy hubs (Bornapour et al., 2016; Bahmani et al., 2021; Benyaghoob-Sani et al., 2021; Heidari et al., 2020; Najafi-Ghalelou et al., 2019; Wen et al., 2022; Brahman et al., 2015; Wang et al., 2018), supermarket refrigeration systems (Brok et al., 2022; Hovgaard et al., 2012b,a; Weerts et al., 2014; Shafiei et al., 2015; Pedersen et al., 2017), buildings (Hajiah and Krarti, 2012a,b; Tang and Wang, 2019; Hu et al., 2019; Arteconi et al., 2017; Doosti et al., 2022), and food industry (Chen et al., 2019; Giordano et al., 2023; Pazmiño-Arias et al., 2023; Cirocco et al., 2022; Saffari et al., 2018).

In most cases, energy hubs are based on fixed load profiles neglecting the system dynamics. Their goal is to optimize the efficient supply of energy. To provide flexibility for load shifting, active energy storages are used. Due to neglecting the system dynamics, load profiles (being dependent on the system state) or thermal inertia of the system (being a passive TES) cannot be investigated using these methods. While Brok et al. (2022) use an active TES as flexibility in a supermarket, other publications considering supermarkets (Hovgaard et al., 2012b,a; Weerts et al., 2014; Shafiei et al., 2015; Pedersen et al., 2017) show the potential of the thermal mass of food as passive TES. It is shown in Hajiah and Krarti (2012a,b), Tang and Wang (2019), Hu et al. (2019), Arteconi et al. (2017) and Doosti et al. (2022) that the thermal mass of buildings can also be used as passive TES for DR.

The available literature on DR in the food industry (Chen et al., 2019; Giordano et al., 2023; Pazmiño-Arias et al., 2023; Cirocco et al., 2022; Saffari et al., 2018) uses active TESs as flexibility. Chen et al. (2019) modeled an energy hub to simulate an industrial park with an electrical power demand for food processing and heat power demand for residential hot water supply. Historical load profiles were used. The optimization is a two-stage robust planning-operation co-optimization based on mixed integer linear programming (MILP) to minimize the electricity costs in a time-of-use (TOU) scenario. As flexibility, a battery storage system and an active TES were considered. Giordano et al.

Table 1

Summary of the literature review. Additional abbreviations used in the table are: nonlinear programming (NLP); cost (C); energy (E); peak load (PL); demand response events (DE) real-time pricing (RTP).

Paper	DSM category	Sector	System dynamics	Production process	opt. method	opt. obj.	Price	Active TES	Passive TES	Uncer-tainty
Nacer et al. (2016a)	EE	Dairy						✓		
Nacer et al. (2016b)	EE	Dairy (milk)								
Maammeur et al. (2017)	EE	Dairy (milk)								
Houston et al. (2014)	EE	Dairy (cheese)								
Philipp et al. (2018)	EE	Dairy (milk)	✓	✓				✓		
Briceño-León et al. (2021)	EE	Bakery								
Becker et al. (2012)	EE	Dairy (cheese)	✓	✓						
Bornapour et al. (2016)	DR	General load			MINLP	C	RTP	✓		✓
Bahmani et al. (2021)	DR	General load			MILP	C	RTP	✓		
Benyaghoob-Sani et al. (2021)	DR	General load			MILP	C, E	RTP	✓		✓
Heidari et al. (2020)	DR	General load			MINLP	C	RTP	✓		✓
Najafi-Ghalelou et al. (2019)	DR	General load			MILP	C, E	RTP, TOU	✓		✓
Wen et al. (2022)	DR	General load			Metaheuristic	C	RTP	✓		✓
Brahman et al. (2015)	DR	resid. load	✓		MILP	C, E	TOU	✓	Air	
Wang et al. (2018)	DR	resid. load			MILP	C	RTP	✓		
Brok et al. (2022)	DR	Supermarket	✓		Numerical	C	RTP	✓		✓
Hovgaard et al. (2012b)	DR	Supermarket	✓		LP, NLP	C	RTP		Food	
Hovgaard et al. (2012a)	DR	Supermarket	✓		NLP	C	RTP		Food	✓
Weerts et al. (2014)	DR	Supermarket	✓		MILP	C	RTP		Food	
Shafiei et al. (2015)	DR	Supermarket	✓		Data-driven	DE			Food	
Pedersen et al. (2017)	DR	Supermarket	✓						Food	
Hajiah and Krarti (2012a)	DR	comm. building	✓		NLP	C	TOU	✓	Building	
Hajiah and Krarti (2012b)	DR	comm. building	✓		NLP	C	TOU	✓	Building	
Tang and Wang (2019)	DR	comm. building	✓		LP	DE			Building	✓
Hu et al. (2019)	DR	resid. building	✓		MILP	C	RTP		Building	✓
Arteconi et al. (2017)	DR	ind. building	✓		Rule-based	C	TOU	✓		
Doosti et al. (2022)	DR	ind. building			MILP	C	RTP	✓		✓
Chen et al. (2019)	DR	Food			MILP	C	TOU	✓		✓
Giordano et al. (2023)	DR	Dairy (milk)	✓		Rule-based	E		✓		
Pazmiño-Arias et al. (2023)	DR	Dairy			MINLP	C	TOU	✓		
Cirocco et al. (2022)	DR	Winery	✓		NLP	RTP	C	✓		✓
Saffari et al. (2018)	DR	Dairy(cheese)			MILP, SP	C	TOU	✓		✓
This paper	DR	Dairy (cheese)	✓	✓	LP	C, E, PL	RTP	✓	Building	✓

(2023) investigated the energy supply of a food processing plant, where milk is the main product. Fixed load profiles were used for the production process. The energy supply was modeled in detail and active energy storages were considered to provide flexibility. The energy consumption was optimized via a rule-based operation leading to reduced energy demand and carbon emissions. Pazmiño-Arias et al. (2023) optimized the energy supply of a dairy factory based on an energy hub, representing the dairy factory by fixed load profiles for heat, cooling, and electricity. The energy hub was optimized via mixed integer nonlinear programming (MINLP) minimizing the total energy costs in a TOU price scenario. Cirocco et al. (2022) investigated the DSM potential of an industrial TES combined with a photovoltaic system (PV) in a winery in Australia. Historical load profiles were used to predict future load profiles. Subsequently, a real TES was used to increase the self-consumption of the PV. Considering an additional PV installation, a nonlinear optimal control algorithm led to electricity cost savings of 22% compared to a non-optimized reference case without PV. Saffari et al. (2018) conducted a case study in a dairy factory, where yogurt and cheese are produced and cooled for storage. The combination of a TES and a PV is investigated, using a MILP formulation based on load profiles for optimization. Simulations show a cost reduction of 1.5% to 10% depending on the scenario.

To the best of the author's knowledge, linear modeling with piece-wise constant parameters in a state space representation has not yet been combined with linear programming (LP) or MILP and, in turn, applied to the industrial food sector. LP (Hovgaard et al., 2012b; Tang and Wang, 2019) or MILP (Bahmani et al., 2021; Benyaghoob-Sani et al., 2021; Najafi-Ghalelou et al., 2019; Brahman et al., 2015; Wang et al., 2018; Weerts et al., 2014; Hu et al., 2019; Doosti et al., 2022; Chen et al., 2019; Saffari et al., 2018) are the preferred choices in literature for optimization, as both provide the global solution and are computationally efficient (Boyd and Vandenberghe, 2004). State space is a formulation often used in control theory and allows the application of well-known methods for modeling the system's behavior, as shown in Tang and Wang (2019) and Hu et al. (2019). Also, no linear model of a food processing plant including the food production process and system dynamics was presented so far in literature. In contrast to commercial building models, as presented by Tang and Wang (2019) and Hu et al. (2019), to successfully apply DR in food production, the process steps need to be modeled due to their significant impact on the system's thermal behavior. Available DR literature in the food industry only investigates active energy storages as flexibility (Chen et al., 2019; Giordano et al., 2023; Pazmiño-Arias et al., 2023; Cirocco et al., 2022; Saffari et al., 2018), is based on fixed load profiles neglecting the building's system dynamics (Chen et al., 2019; Pazmiño-Arias et al., 2023; Saffari et al., 2018), is solved rule-based (Giordano et al., 2023), or solved based on nonlinear optimization techniques (Cirocco et al., 2022).

In this work, a case study of an Austrian food processing plant is considered, and the DR potential of the industrial refrigeration system in combination with a chilled water buffer as active and building mass as passive TES is investigated. The main contributions are:

- We propose a model of a food processing plant based on transient energy balances. The model is formulated in state space and is linear with piece-wise constant parameters to allow for the application of LP techniques, leading to a computationally efficient algorithm and a global optimum.
- We investigate two settings, where (1) the thermal mass of the building as passive TES and (2) a chilled water buffer as active TES are used as flexibility for DSM. Exploiting these load-shifting potentials, we optimize the total electrical energy consumption, the total electrical energy costs, and the electrical peak load of the plant separately.
- To handle prediction uncertainties and disturbances, we implement the optimization problems within a model predictive control (MPC) setting.

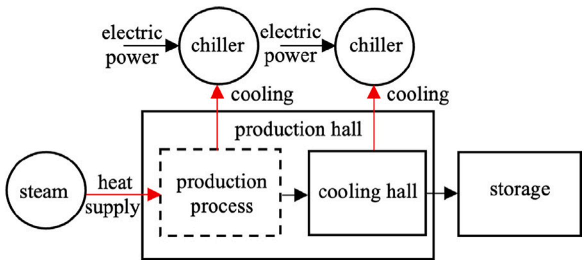

Fig. 1. Simplified scheme of the processed food plant showing the most important components and process steps.

- We conduct a case study of a food processing plant including system identification and simulations. Via sensitivity analysis, varying sizes of the active TES for DR are evaluated.

The paper is organized as follows: The system model, the optimization problem, and the MPC algorithm are described in Section 2. In Section 3, we identify the system parameters and present the optimization results including the sensitivity analysis. Finally, conclusions are drawn in Section 4.

2. Methods

Fig. 1 shows a simplified sketch of the food processing plant considered. Various cheese products are produced in about 20 different production lines. During the production process, the food product is heated first, later cooled down, and finally stored. The heat necessary for the process is provided by a steam boiler and two chillers are used to cool the production hall and the food products in the cooling hall. In Section 2.1, we formulate a model describing the production process with only limited input data. The production plan, the ambient temperature, and the solar radiation are assumed to be known comprising the input data of the model. The model is considered to be capable of representing the system dynamics sufficiently for subsequent analyses.

2.1. Model of the plant

Fig. 2 shows a more detailed scheme of the food processing plant including the heat transfers considered. The batch process is modeled in $N_\mathrm{F} = 18$ steps using the index n to differentiate the single process steps. At step $n = 1$, the product is heated to approximately 90 °C using steam and served ingredients are mixed in. At step $n = 2$, the fluid products are poured into molds and transported to the cooling room, where the products are cooled down to approximately 5 °C for storage. The cooling process is divided into several steps so that every process step can be represented by a step of equal duration Δt. The building is divided into three subparts: the production hall, the cooling hall, and the thermal mass of the building envelope. The system temperatures depend on the heat loss of the production lines, the outside temperature, the solar radiation, and the heat extracted by the two chillers.

The behavior of the steam boiler was analyzed and described in detail in previous work (Wohlgenannt et al., 2022), accordingly, we use the following simplified model:

$$\dot{Q}_\mathrm{S} = \eta_\mathrm{G} P_\mathrm{G}, \tag{1}$$

$$\dot{Q}_\mathrm{S,F} = \eta_\mathrm{F} \dot{Q}_\mathrm{S}, \tag{2}$$

$$\dot{Q}_\mathrm{S,W} = (1 - \eta_\mathrm{F}) \dot{Q}_\mathrm{S}, \tag{3}$$

where P_G is the gas power used, η_G is the efficiency of the steam boiler, $\dot{Q}_\mathrm{S}$ is the resulting steam heat transfer rate, and η_F is the efficiency of the heating process. $\dot{Q}_\mathrm{S,F}$ refers to the steam that heats the food product and $\dot{Q}_\mathrm{S,W}$ is the waste heat that heats the production hall. The cooling powers of the chillers ($\dot{Q}_\mathrm{Ch,1}$ and $\dot{Q}_\mathrm{Ch,2}$) are calculated by using the electrical powers ($P_\mathrm{El,1}$ and $P_\mathrm{El,2}$) and energy efficiency ratios (EERs) (ε_1 and ε_2), which are linearly dependent (modeled via a and b) on the ambient temperature T_∞ as follows:

$$\dot{Q}_\mathrm{Ch,1}(t) = P_\mathrm{El,1}(t)\varepsilon_1(t), \tag{4}$$

$$\dot{Q}_\mathrm{Ch,2}(t) = P_\mathrm{El,2}(t)\varepsilon_2(t), \tag{5}$$

$$\varepsilon_1(t) = a_1 T_\infty(t) + b_1, \tag{6}$$

$$\varepsilon_2(t) = a_2 T_\infty(t) + b_2. \tag{7}$$

The influence of the sun is modeled via the absorbed solar radiation ($\alpha \dot{Q}_\mathrm{Sun}$), which is assumed to heat the building envelope. The heat transfer rates from the building envelope to the production hall and the surroundings are modeled via single heat transfer characteristics (k) to reflect convection. The equations representing the heat transfer rates are given by:

$$\dot{Q}_\mathrm{B2\infty}(t) = k_\mathrm{B2\infty}\left(T_\mathrm{B}(t) - T_\infty(t)\right), \tag{8}$$

$$\dot{Q}_\mathrm{B2P}(t) = k_\mathrm{B2P}\left(T_\mathrm{B}(t) - T_\mathrm{P}(t)\right), \tag{9}$$

$$\dot{Q}_\mathrm{P2C}(t) = k_\mathrm{P2C}\left(T_\mathrm{P}(t) - T_\mathrm{C}(t)\right), \tag{10}$$

$$\dot{Q}_\mathrm{F,1}(t) = k_\mathrm{F,1}(t)\left(T_\mathrm{F,1}(t) - T_\mathrm{P}(t)\right) + \dot{Q}_\mathrm{S,W}, \tag{11}$$

$$\dot{Q}_\mathrm{F,2}(t) = k_\mathrm{F,2}(t)\left(T_\mathrm{F,2}(t) - T_\mathrm{P}(t)\right), \tag{12}$$

$$\dot{Q}_{\mathrm{F},n}(t) = k_{\mathrm{F},n}(t)\left(T_{\mathrm{F},n}(t) - T_\mathrm{C}(t)\right) \text{ for } 3 \le n \le N_\mathrm{F}, \tag{13}$$

where $\dot{Q}_\mathrm{B2\infty}$ is the heat transfer rate from the building envelope to ambient, $\dot{Q}_\mathrm{B2P}$ is the heat transfer rate from the building envelope to the production hall, $\dot{Q}_\mathrm{P2C}$ is the heat transfer rate from the production hall to the cooling hall, and $\dot{Q}_{\mathrm{F},n}$ is the heat transfer rate from the food product to the production or cooling hall. The waste heat transfer rate of the food product heating process at process step $n = 1$ is added to $\dot{Q}_\mathrm{F,1}$. Note that while $k_\mathrm{B2\infty}$, k_B2P, and k_P2C are assumed to be constant, $k_{\mathrm{F},n}$ depends on the amount of food produced per interval of the batch process ($m_{\mathrm{F},n}$). This dependency is assumed to be linear because the specific heat transfer characteristics are assumed to be constant and the surface of the produced food is assumed to be proportional to the amount of food produced. The energy balances of Fig. 2 for one interval of the batch process can be written as:

$$\dot{T}_\mathrm{B}(t) = \frac{1}{C_\mathrm{B}}\Big[\alpha \dot{Q}_\mathrm{Sun}(t) + k_\mathrm{B2\infty}\left(T_\infty(t) - T_\mathrm{B}(t)\right) + k_\mathrm{B2P}\left(T_\mathrm{P}(t) - T_\mathrm{B}(t)\right)\Big], \tag{14}$$

$$\dot{T}_\mathrm{P}(t) = \frac{1}{C_\mathrm{P}}\Big[\dot{Q}_\mathrm{S,W}(t) - \dot{Q}_\mathrm{Ch,1}(t) + k_\mathrm{B2P}\left(T_\mathrm{B}(t) - T_\mathrm{P}(t)\right) + k_\mathrm{P2C}\left(T_\mathrm{C}(t) - T_\mathrm{P}(t)\right) + \sum_{n=1}^{2} k_{\mathrm{F},n}\left(T_{\mathrm{F},n}(t) - T_\mathrm{P}(t)\right)\Big], \tag{15}$$

$$\dot{T}_\mathrm{C}(t) = \frac{1}{C_\mathrm{C}}\Big[-\dot{Q}_\mathrm{Ch,2}(t) + k_\mathrm{P2C}\left(T_\mathrm{P}(t) - T_\mathrm{C}(t)\right) + \sum_{n=3}^{N_\mathrm{F}} k_{\mathrm{F},n}\left(T_{\mathrm{F},n}(t) - T_\mathrm{P}(t)\right)\Big], \tag{16}$$

$$\dot{T}_\mathrm{F,1}(t) = \frac{1}{C_\mathrm{F,1}(t)}\Big[\dot{Q}_\mathrm{S,F}(t) + k_\mathrm{F,1}(t)\left(T_\mathrm{P}(t) - T_\mathrm{F,1}(t)\right)\Big], \tag{17}$$

$$\dot{T}_\mathrm{F,2}(t) = \frac{1}{C_\mathrm{F,2}(t)} k_\mathrm{F,2}(t)\left(T_\mathrm{P}(t) - T_\mathrm{F,2}(t)\right), \tag{18}$$

$$\dot{T}_{\mathrm{F},n}(t) = \frac{1}{C_{\mathrm{F},n}(t)} k_{\mathrm{F},n}(t)\left(T_\mathrm{C}(t) - T_\mathrm{F,n}(t)\right) \text{ for } 3 \le n \le N_\mathrm{F}, \tag{19}$$

where C refers to thermal capacity. Assuming $\dot{Q}_\mathrm{Sun}(t)$, $\dot{Q}_\mathrm{Ch,1}(t)$, $\dot{Q}_\mathrm{Ch,2}(t)$, $\dot{Q}_\mathrm{S,P}(t)$, $\dot{Q}_\mathrm{S,F}(t)$, $T_\infty(t)$, $k_{\mathrm{F},n}(t)$, and $C_{\mathrm{F},n}(t)$ to be constant in each time interval of the batch process, this defines a system of coupled first-order inhomogeneous linear ordinary differential equations (ODEs) with constant coefficients. Constant $C_{\mathrm{F},n}$ reflect the assumption that the

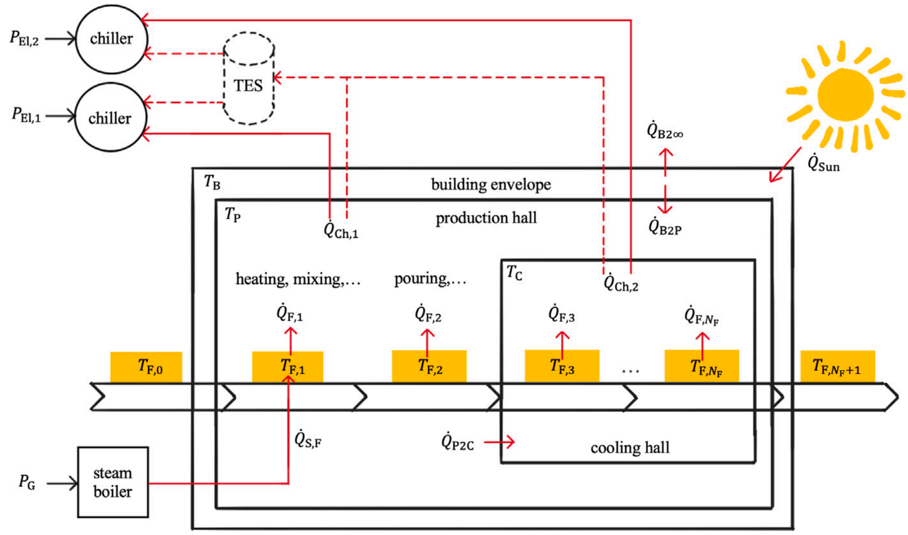

Fig. 2. Scheme of the processed food plant showing all relevant heat transfers. The dashed lines show a potential extension of the existing plant considered by simulation. The extension considers a refrigeration network combining both chillers, extended by an additional TES.

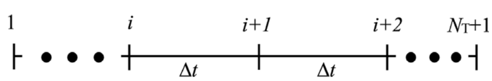

Fig. 3. Indices used for discretization, considering N_T time steps, each of duration Δt resulting in $N_T + 1$ points in time.

food is produced in a batch process and the mass is transported from one process step n to the next one in the transition between the time intervals. The system of ODEs can be given in the following form:

$$\dot{\boldsymbol{T}}(t) = \boldsymbol{A} \cdot \boldsymbol{T}(t) + \boldsymbol{b} + \boldsymbol{u}, \forall t \in [t^{(i)}, t^{(i+1)}], \tag{20}$$

where $\boldsymbol{A}$ is the coefficient matrix, $(\boldsymbol{b} + \boldsymbol{u})$ is the inhomogeneous part, and $\boldsymbol{u}$ is the control vector later being used as the decision variable of the optimization problems. These ODEs define the state space of the system considered and can be solved using the matrix exponential and the steady state solution $\boldsymbol{T}_{ss}$ as follows:

$$\boldsymbol{T}_{ss} = -\boldsymbol{A}^{-1}(\boldsymbol{b} + \boldsymbol{u}) \tag{21}$$

$$\boldsymbol{T}(t) = e^{\boldsymbol{A}t}[\boldsymbol{T}_0 - \boldsymbol{T}_{ss}] + \boldsymbol{T}_{ss} \tag{22}$$

This solution can be applied for piece-wise constant parameters on N_T intervals of duration Δt. The indexing of the time discretization is shown in Fig. 3. This results in

$$\boldsymbol{T}^{(i+1)} = e^{\boldsymbol{A}\Delta t}[\boldsymbol{T}^{(i)} - \boldsymbol{T}_{ss}] + \boldsymbol{T}_{ss}. \tag{23}$$

Using this analytical solution allows for the calculation of the system dynamics at all time points indexed by $i = 1, \ldots, i = N_T + 1$. Note that the food is produced in a batch process, and, therefore the final temperature $T_{F,n}(t + \Delta t)$ at the end of the time interval is the start value for the temperature $T_{F,n+1}(t = 0)$ at the next process step.

2.2. Chilled water buffer

Additionally, a chilled water buffer is added in simulations as active TES. Both chillers are using the same TES (as seen dashed in Fig. 2). They are combined to a refrigeration network such that each chiller can cool both halls. The active TES is modeled via

$$E^{(i+1)} = E^{(i)} + \dot{Q}_{Ch,1}\Delta t + \dot{Q}_{Ch,2}\Delta t - \dot{Q}_{dem}\Delta t, \tag{24}$$

$$E_{max} = c_E \rho V_{max} \Delta T_E, \tag{25}$$

$$SOC^{(i)} = \frac{E^{(i)}}{E_{max}} \tag{26}$$

where $E^{(i)}$ is the energy stored in the TES, SOC is the state of charge of the TES, $\dot{Q}_{dem}$ is the cooling demand, E_{max} is the maximum energy, V_{max} is the volume of the storage, c_E and ρ are the specific thermal capacity and the density of the cooling medium, respectively, and ΔT_E refers to the maximum temperature difference considered in the storage.

2.3. P-controllers

In the real plant, the chillers are controlled via two digital P-controllers. In the remainder of the paper, this setting is considered the reference case all optimizations are compared to. The following equations describe the controllers:

$$\dot{Q}_{Ch,1} = K_{P,1}(T_P - T_{P,set}) \tag{27}$$

$$\dot{Q}_{Ch,2} = K_{P,2}(T_C - T_{C,set}) \tag{28}$$

The cycle time of the digital P-controllers is assumed to be 1 min, the set point of the production hall is $T_{P,set} = 20$ °C, the set point of the cooling hall is $T_{C,set} = 0$ °C, and the proportional factors are $K_{P,1} = 100000$ W/K and $K_{P,2} = 500000$ W/K.

2.4. Optimization problems

The objective is to reduce energy consumption, energy costs, or peak load by controlling both chillers. Under the dynamic constraints presented above, the optimization problem that minimizes the total energy consumption is given by

$$\min_{P_{El,1}, P_{El,2}} \sum_{i=1}^{N_T} (P_{El,1}^{(i)} + P_{El,2}^{(i)})\Delta t. \tag{29}$$

The optimization problem that minimizes the total energy costs is given by

$$\min_{P_{El,1}, P_{El,2}} \sum_{i=1}^{N_T} (P_{El,1}^{(i)} + P_{El,2}^{(i)})\Delta t \pi^{(i)}, \tag{30}$$

where $\pi^{(i)}$ is the energy price for time interval i. The peak load reduction problem is given by

$$\min_{P_{El,1}, P_{El,2}} \max_{i=1,\ldots,N_T} (P_{El,1}^{(i)} + P_{El,2}^{(i)}). \tag{31}$$

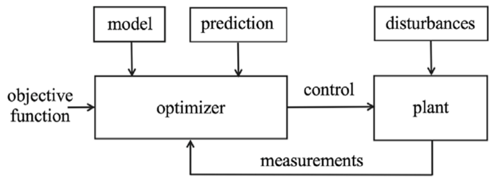

Fig. 4. Model predictive control concept.

The objective function in Eq. (31) is nonlinear, however, it can be rewritten in a linear form (Linear, 2023) by introducing an auxiliary variable a resulting in:

$$\min_{P_{\mathrm{El},1},P_{\mathrm{El},2}} \quad a, \tag{32}$$

$$\text{s.t. } a \geq (P_{\mathrm{El},1}^{(i)} + P_{\mathrm{El},2}^{(i)}). \tag{33}$$

The following constraints limit the range of the decision variables $P_{\mathrm{El},1}$ and $P_{\mathrm{El},2}$ $\forall i \in [1, \ldots, N_{\mathrm{T}}]$:

$$0 \leq P_{\mathrm{El},1}^{(i)} \leq P_{\mathrm{El},1,\max}, \tag{34}$$

$$0 \leq P_{\mathrm{El},2}^{(i)} \leq P_{\mathrm{El},2,\max}. \tag{35}$$

2.4.1. Passive energy storage

In this setting, the thermal mass of the production and cooling halls are used as passive energy storage. A temperature band for both halls is defined to provide flexibility for load management. The constraints are:

$$T_{\mathrm{P,min}} \leq T_{\mathrm{P}}^{(i+1)} \leq T_{\mathrm{P,max}}, \tag{36}$$

$$T_{\mathrm{C,min}} \leq T_{\mathrm{C}}^{(i+1)} \leq T_{\mathrm{C,max}}, \tag{37}$$

where T_{P} and T_{C} according to Eqs. (14)–(19) represent the building and Eqs. (36)–(37) define the allowed temperature ranges of the production and the cooling hall.

2.4.2. Active energy storage

In this setting, the chilled water buffer is used as active energy storage. The constraints to reflect the lower and upper boundaries of the TES are

$$E_{\min} \leq E^{(i+1)} \leq E_{\max}. \tag{38}$$

2.5. Model predictive control

The MPC concept is shown in Fig. 4. In MPC, the model behavior is predicted up to a certain time horizon. Usually, only the control signal for the next time step is used, and then the optimization is repeated. Thereby, the optimization is performed in the next time step according to the updated predictions and the updated measured state. This can be understood as a feedback loop and makes MPC a closed-loop control in contrast to optimal control (Camacho and Bordons, 2007; Grüne and Pannek, 2017). By repeating the optimization with new information, the optimizer is capable of handling uncertainties and disturbances by adapting to evolving system conditions. The plant model defines the constraints of the optimization problem. Predictions of the uncertain inputs ($m_{P,n}$, $\dot{Q}_{\mathrm{S}}$, $\dot{Q}_{\mathrm{Sun}}$, T_{∞}, $\dot{Q}_{\mathrm{dem}}$) of the optimization are used.

The three objectives are investigated: (1) energy optimization, (2) cost optimization, and (3) peak load reduction. All are compared with the reference case, where two digital P-controllers are used for the chillers. While for (1) and (3), the optimization window is considered to be 48 h, the optimization window for (2) is not constant and depends on how long the day-ahead electricity price is considered to be known in advance. Here, the Energy Exchange Austria (EXAA) (APG, 2023) spot market price is used for RTP. At noon, the prices for the next day are published, resulting in an optimization window between 36 h and 12 h. Fig. 5 shows the spot market price for an example day.

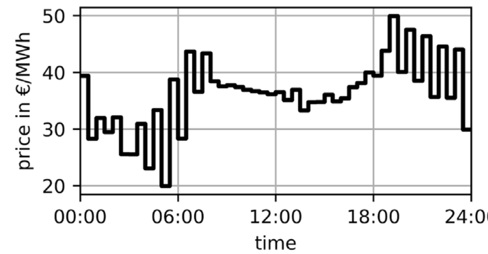

Fig. 5. Spot market price for an example day (13 July 2020).

2.6. Predictions

Uncertain inputs of the optimization ($m_{F,n}$, $\dot{Q}_{\mathrm{S}}$, $\dot{Q}_{\mathrm{Sun}}$, T_{∞}, $\dot{Q}_{\mathrm{dem}}$) need to be predicted. For prediction, 2 scenarios are evaluated: (1) perfect prediction and (2) realistic prediction. In perfect prediction, we assume the uncertain inputs to be perfectly known in advance. In the realistic prediction scenario, a simple prediction approach based on averages of historical data and measured data from the last 30 min is used. Due to the system's high inertia, we consider measured values from the last 30 min as predictions for the next 30 min. The remaining values are predicted via historical data. The uncertain inputs are split into production-dependent and production-independent inputs. $\dot{Q}_{\mathrm{Sun}}$ and T_{∞} are independent of the production and, therefore, the complete training data can be used to calculate an average day. In contrast, $m_{P,n}$, $\dot{Q}_{\mathrm{S}}$, $\dot{Q}_{\mathrm{dem}}$ are dependent on the production and, therefore, an average weekday, an average Saturday, and an average Sunday are calculated. The differentiation between Saturday and Sunday is necessary because, at the factory investigated, production on weekends is reduced significantly and even stopped completely on Sunday. The production and the resulting load profiles during the weekdays are very similar, therefore, one prediction for a weekday profile is considered sufficient.

2.7. Applicability to similar problems

The modeling approach chosen can be applied to thermal systems with similar constraints. In this work, the constraints considered are: (1) no mass flow in the energy balance during time intervals of a fixed size Δt, i.e., mass flow only occurs in the transition between the time intervals; (2) piece-wise constant parameters and inputs such that the coefficient matrix ($\boldsymbol{A}$) and the inhomogeneous part ($\boldsymbol{b} + \boldsymbol{u}$) of the underlying system of ODEs are constant for each time interval. This modeling approach results in a linear state space representation with piece-wise linear parameters.

The applied optimization approach can be applied to any system (electrical, thermal, mechanical, …) in state space representation if (3) the decision variables of the optimization problem are only present in the inhomogeneous part ($\boldsymbol{b} + \boldsymbol{u}$) of the system's dynamics.

3. Results and discussion

3.1. System identification

Historical data from July 2020 is available. The electrical power consumption of both chillers ($P_{\mathrm{El},1}$ and $P_{\mathrm{El},2}$), the cooling power of one chiller ($\dot{Q}_{\mathrm{Ch},2}$), the process heat transfer rate $\dot{Q}_{\mathrm{S}}$, the product temperature ($T_{\mathrm{F},n}$), and the production quantity per day are known. For the outside temperature (T_{∞}) and the solar radiation ($\dot{Q}_{\mathrm{Sun}}$), data from a local weather station is used (DWD, 2023). The range of the specific thermal heat coefficient ($c_{\mathrm{F},n}$) of the product can be taken from literature Pajonk et al. (2003) and for discretization, a time step of $\Delta t = 30$ min is used.

For a fully parameterized model following parameters are missing and need to be identified: EER of the two chillers (a_1, b_1, a_2 and b_2), the

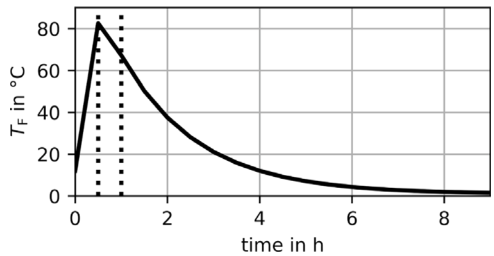

Fig. 6. Temperature profile of food product during the production process. The vertical, dotted lines indicate the different steps during the process, starting with the heating and mixing, continuing with the pouring, and, finally, the cooling.

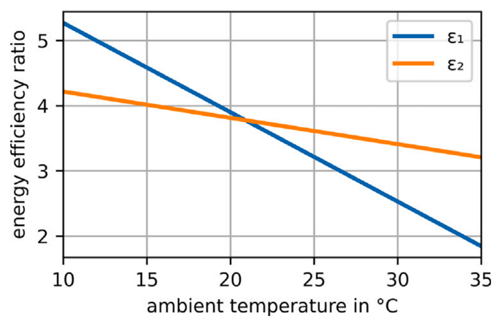

Fig. 7. Energy efficiency ratio of both chillers in dependence of the ambient temperature.

thermal mass of the building (C_B), the thermal mass of the production hall (C_P), the thermal mass of the cooling hall (C_C), the heat transfer coefficients ($k_{B2\infty}$, k_{B2P}, k_{P2C} and $k_{F,n}$), the absorptance rate (α), and the efficiency of the heating process (η_F).

The system identification is down in two stages. In stage one, system components with additional information available are estimated separately. In stage two, the remaining parameters are fitted via one least square estimation problem.

In stage one, the EER (a_2, b_2) of chiller 2 can be estimated via least squares using measured data of $P_{El,2}$ and $Q_{Ch,2}$. Also, the temperature profile of the food product during the production process is known and can be seen in Fig. 6. This temperature profile can be used to calculate the heat transfer coefficients of the product ($k_{P,n}$) and the production steam rate $\xi_{S,P}$.

In stage two, the remaining parameters (a_1, b_1, C_P, C_C, C_B, $k_{B2\infty}$, k_{B2P}, k_{P2C}, α, η_F) are fitted via one least square estimation problem using the scipy package (SciPy, 2023). As an error function, the quadratic error of the measured and the estimated electrical power consumption of both chillers is minimized via

$$\min \sum_{i=1}^{N_T} \Big((P_{El,1}^{(i)} - P_{El,1,\ estimated}^{(i)})^2 + (P_{El,2}^{(i)} - P_{El,2,\ estimated}^{(i)})^2 \Big). \quad (39)$$

The results of the parameter fit are shown in Table 2, Figs. 7 and 8.

The R^2 scores of the parameter estimation for $P_{El,1}$ and $P_{El,2}$ are 0.70 and 0.85, respectively. The score of $P_{El,2}$ is significantly better than the score of $P_{El,1}$ because the disturbances such as the ambient temperature or the solar radiation are less influential in case of the cooling hall.

3.2. Realistic predictions

To calculate the realistic prediction, the available data was split into 7 days for training and 24 days for testing. The 7 training days were used to separately predict $m_{F,n}$, $\dot{Q}_S$, $\dot{Q}_{Sun}$, T_∞ and $\dot{Q}_{dem}$ and Table 3 shows the R^2 scores of these predictions.

Table 2
Estimated model parameters.

Parameter	Value
a_1	−0.1638 (1/K)
b_1	51.6080
a_2	−0.0404 (1/K)
b_2	15.6422
C_P	$3.15 \cdot 10^8$ (J/K)
C_C	$1.05 \cdot 10^8$ (J/K)
C_B	$3.57 \cdot 10^9$ (J/K)
$c_{F,n}$	3270 (J/(kg K))
$k_{B2\infty}$	$1.86 \cdot 10^4$ (W/K)
k_{B2P}	$1.28 \cdot 10^4$ (W/K)
k_{P2C}	$2.46 \cdot 10^3$ (W/K)
$k_{F,n}/m_{F,n}$	0.54 (W/(kg K))
α	0.79
η_F	0.95

Table 3
R^2 scores of the predictions.

Parameter	R^2 score
$m_{F,1}$	0.77
$\dot{Q}_{Sun}$	0.73
T_∞	0.52
$\dot{Q}_S$	0.77
$\dot{Q}_{dem}$	0.82

While $m_{F,n}$, $\dot{Q}_S$, $\dot{Q}_{Sun}$ and $\dot{Q}_{dem}$ can be predicted from the limited historical data with decent precision, the prediction of T_∞ is not very precise with an R^2 score of 0.52. However, the prediction of T_∞ is still good enough for the proposed methods, because the basic weather trends are very robust, for example, the average temperature during the night being lower than the average temperature during the day.

3.3. Optimization results

The optimization problems are implemented using Python and Gurobi (Gurobi Optimization, LLC, 2023). As passive energy storage, the thermal capacities of the production hall and the cooling hall were used, allowing a temperature band of 7 K. The production hall temperature can vary between 19 °C and 26 °C and the cooling hall temperature can vary between −3 °C and 4 °C. Alternatively, as active energy storage a chilled water buffer with a size of 100 m^3 is used. The energetic flexibility of a TES with a size of 100 m^3 is approximately equivalent to a hall temperature band of 7 K for DSM. The temperature difference between the chiller's inlet and outlet temperatures is 10 K and the specific heat capacitance of the cooling medium is assumed to be 3525 $\frac{J}{kg\ K}$, which is a typical value of a water–glycol mixture with antifreeze of at least −20 °C. Additionally, MPC with perfect prediction is compared with MPC with realistic prediction based on the data available. All optimizations are compared to the reference case, where P-controllers are used for the chillers. The test set of the data contains 24 days, 22 of which were evaluated, as the last two days serve as data for the next 48 h forecast for the optimization. The results of the optimization for the 22 days are shown in Fig. 9, and the details for one example day are shown in Fig. 10.

3.3.1. Passive energy storage

Compared to the reference case, passive energy storage leads to energy savings, cost savings, and a reduction of peak power. In energy optimization, MPC with realistic prediction and MPC with perfect prediction result in an energy reduction of 17% and 18%, respectively. These energy savings result from higher average temperatures and thus lower heat losses to the ambient. So, less cooling power is needed, and the chillers are operated during times with high efficiency. In the cost optimization, MPC with realistic prediction and MPC with perfect prediction result in cost reductions of 22% and 23%, respectively. The

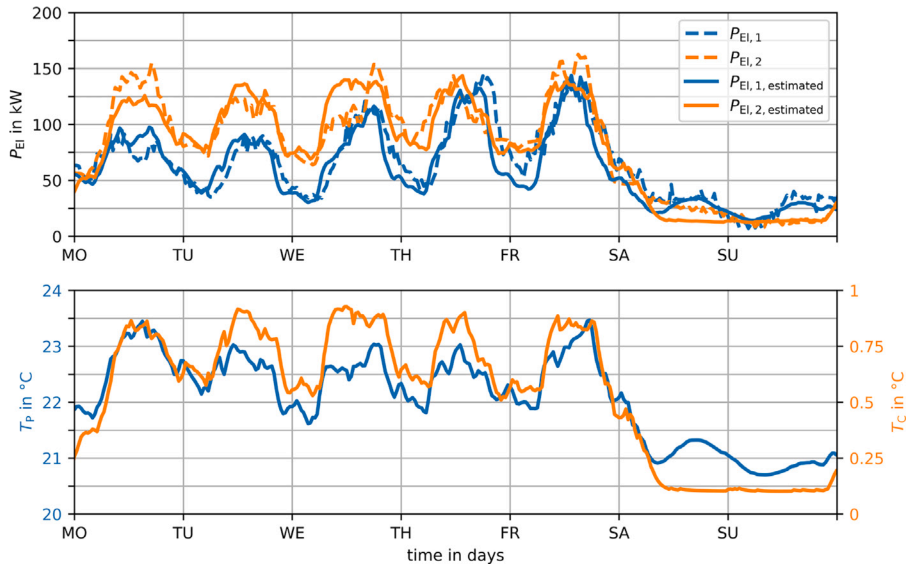

Fig. 8. Comparison of the estimated and measured cooling powers (top), estimated temperature profiles of the production hall, and the cooling hall (bottom) for one week.

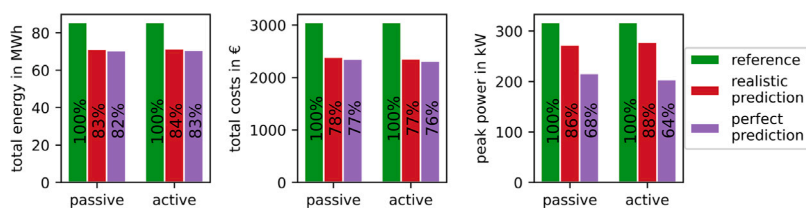

Fig. 9. Total energy, total costs, and peak load of the 22 simulated days. The two settings active and passive TES are evaluated and three objectives are optimized separately: (1) energy consumption, (2) electricity costs, and (3) peak load. MPC with realistic predictions is compared to MPC with perfect prediction and with the reference case of P-controllers.

savings are achieved via load shifting and energy savings. The loads are shifted to times with lower prices and the total energy consumption is reduced. It can be seen in Fig. 10 that in both cases, the prediction quality is not the dominant factor. Thereby, the MPC without perfect prediction resulted in nearly as good results as MPC with perfect prediction. The reasons are the big thermal capacities. Due to the high capacities, the system reacts very slowly to changes; Thereby, using the historical values of the last 30 min as predictions, are proven to be accurate enough for the next time interval. Considering peak load, MPC with realistic and perfect prediction resulted in a peak load reduction of 14% and 32% respectively. The reason for the significant impact of the prediction on the performance is that when the energy storage level is close to a limit, a single prediction error can lead to a new power peak.

3.3.2. Active energy storage

The second setting is DSM with active energy storage, which leads to similar results compared to the passive energy storage. The energy optimization with realistic and perfect prediction reduces the total energy consumption by 16% and 17% respectively. Most of the energy savings from the active energy storage result from the connection of both chillers to a refrigeration system so that both chillers can cool both halls. Thereby, always the chiller with better efficiency can be used, which can be seen in Fig. 10. Fig. 7 shows that chiller 1 is prioritized due to its higher efficiency during low ambient temperatures, while chiller 2 exhibits higher efficiency during high ambient temperatures. The cost optimization with real and perfect prediction reduces the total electricity costs by 23% and 24%, respectively. The peak load is reduced by 12% using realistic predictions and by 36% assuming perfect prediction. Similar to the simulations with the passive energy storage, the prediction quality has a high impact on the performance of the peak load reduction.

Table 4
Execution time of the optimization problems running on a Macbook Pro M2 2022.

TES	Objective	Time
Passive	Energy opt.	1.24 s
Passive	Cost opt.	0.82 s
Passive	Peak load red.	1.18 s
Active	Energy opt.	0.49 s
Active	Cost opt.	0.27 s
Active	Peak load red.	0.44 s

3.3.3. Execution time

All formulations of optimization problems proposed are computationally efficient, exhibiting execution times to initialize and solve an optimization problem between 0.27 s and 1.24 s (see Table 4). The fast execution time is achieved due to the linearity of the formulations.

3.4. Sensitivity analysis

A sensitivity analysis is conducted to evaluate the influence of the energy storage size on the results in both settings. The allowed temperature band of the production and the cooling hall is evaluated from 0 K to 10 K. The buffer size was evaluated from 0 m^3 to 200 m^3, where 0 m^3 means that both chillers are connected and can cool both halls, but there is no buffer. The results of this sensitivity analysis are given in Fig. 11.

With perfect prediction and passive energy storage, the minimum temperature band that satisfies all constraints is 1 K. Without perfect prediction, this minimum temperature band is 2 K. In the reference case, the minimum temperature band is 3 K in the production hall and 1 K in the cooling hall. It can be seen that small temperature bands result in significant energy savings compared to the reference case

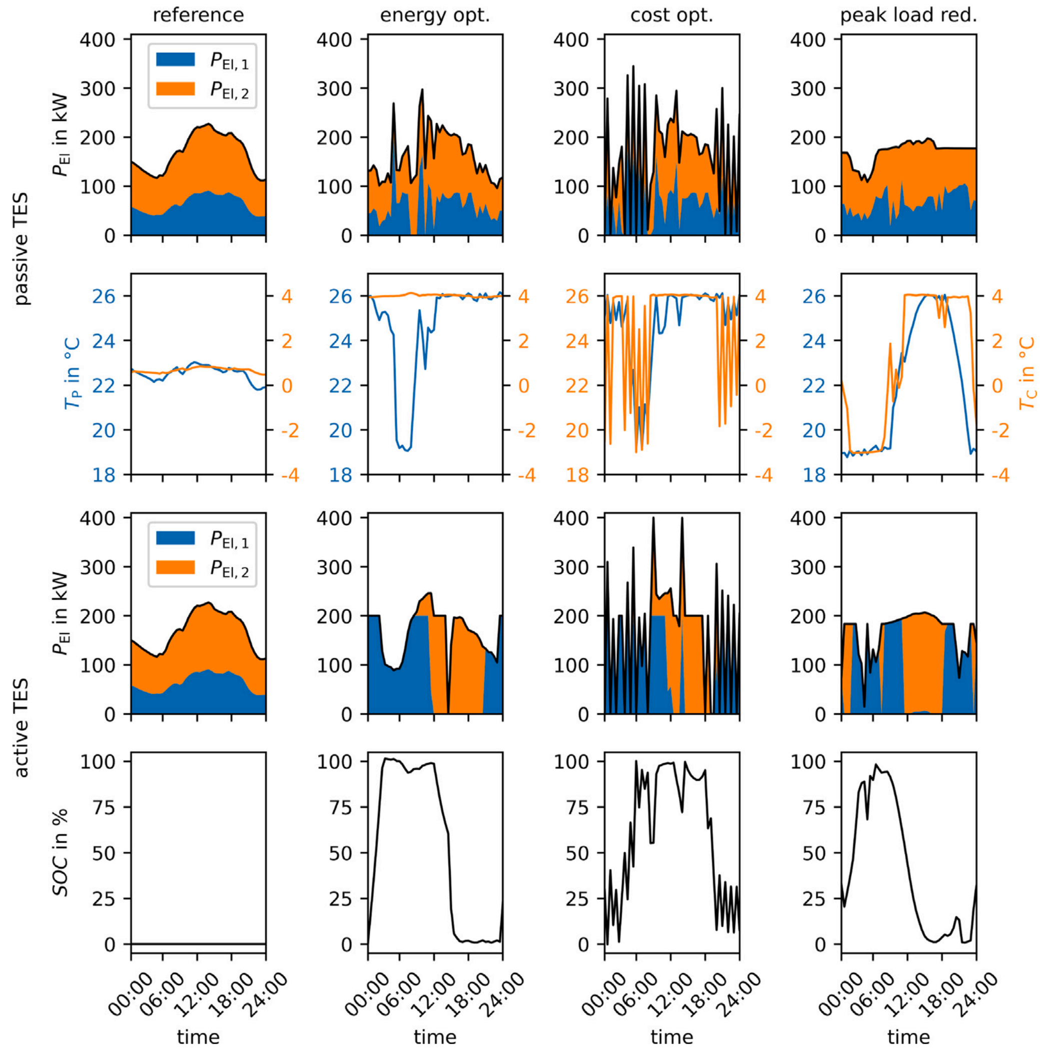

Fig. 10. Optimization results for one example production day (13 July 2020), where MPC with realistic predictions is applied. The two settings active and passive TES are evaluated. The first two rows and the last two rows show the results of the passive and active TES, respectively. The first and the third rows show the electrical power consumption of the chillers, which are depicted by a stack plot. The black line indicates the total electrical power. The second and the fourth rows show the state of the passive and active energy storage. The passive energy storage is represented by the temperatures of the production and the cooling hall and the active energy storage is described via its state of charge (SOC). Three objectives are optimized separately: (1) energy consumption, (2) electricity costs, and (3) peak load and compared with the reference case and are shown in the different columns.

that uses P-controllers. Also, the prediction quality is mainly important in peak load reduction. In the other objectives, the MPC algorithm is able to handle the disturbances without a relevant deterioration. Evaluating the results of the active energy storage it can be seen, that the connection of both chillers to a refrigeration network is the most relevant factor in the energy and cost optimizations. The refrigeration network optimally controls the chiller so that always the chiller with the better EER is used if possible. This reduces energy consumption by 15% and electricity costs by 16%. While there is some potential for a buffer for cost optimization, the potential for energy savings using a buffer are very little. The main benefits of the buffer are for peak load reduction. There, the refrigeration network only reduces the peak by 6% and the remaining reduction is achieved via the chilled water buffer. Also, a clear saturation effect can be seen there.

4. Conclusions

We proposed a model of a food processing plant that can be used for simulation and optimization. The model is based on transient energy balances resulting in a linear state space representation of the system and LP is used for optimization. Due to the properties of LP, the optimizations are computationally efficient. The proposed modeling and optimization approach applies to systems with similar constraints.

Two settings are investigated to provide demand-side flexibility: (1) the thermal mass of the production hall and the cooling hall as passive TES or (2) a chilled water buffer as active TES. The flexibility considered allows to dynamically adapt the control of the two chillers, resulting in a change of electrical power consumption. The optimization objectives are to reduce the total energy consumption, the electricity costs, or the peak load of the plant in three different scenarios considered. The optimization problems are implemented within an MPC algorithm to account for uncertainties and disturbances. The impact of different TES sizes is investigated in a sensitivity analysis.

Within our case study, the energy consumption was reduced by up to 18%, the electricity costs by up to 24%, and the peak power by up to 36%. Comparing perfect to realistic predictions within the MPC approach resulted in similar energy and cost optimization benefits, showing that realistic predictions only performed one percentage point worse on average. On the other hand, the results reflect that an accurate peak load reduction relies on a high prediction quality. Especially in times of the TES being close to a limit, predictions of low accuracy may result in additional power peaks. Considering realistic predictions,

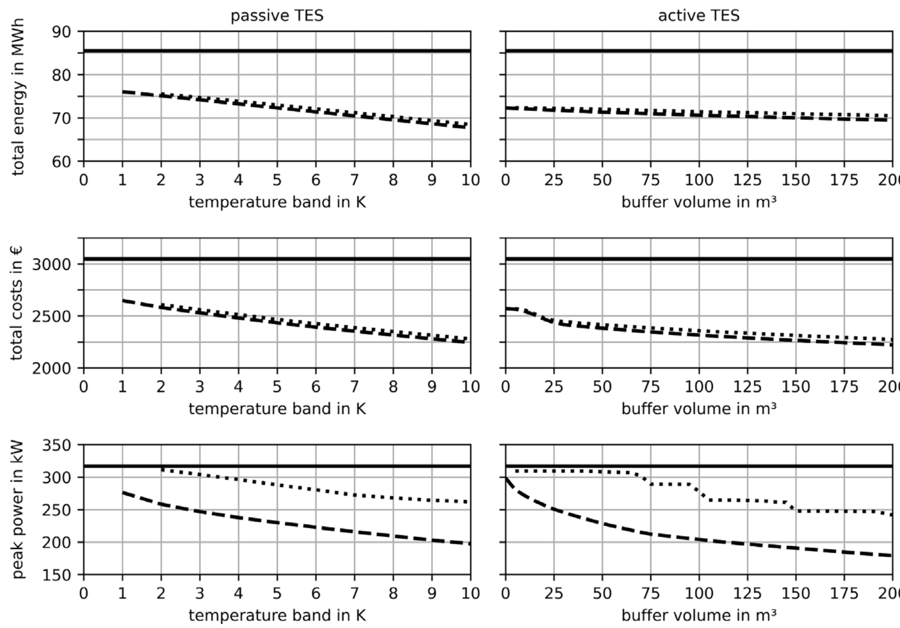

Fig. 11. Sensitivity analysis to evaluate the impact of the TES size on the DSM potential. In the left column passive TES, and in the right column active TES is evaluated. The solid line represents the reference case, the dashed line MPC with perfect prediction, and the dotted line MPC with realistic prediction.

the peak load reduction is less pronounced by up to 66%. Our results show, that applying DSM in food processing plants can save costs and energy, and help stabilize the grid. Therefore, further research in this area is of great interest. Promising further research directions are the application of stochastic programming or robust optimization based on the LP formulation proposed. This could be a viable approach to handle uncertainties and improve the performance in peak load reduction.

CRediT authorship contribution statement

Philipp Wohlgenannt: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Gerhard Huber:** Writing – review & editing, Visualization, Validation, Resources, Methodology, Data curation, Conceptualization. **Klaus Rheinberger:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Mohan Kolhe:** Writing – review & editing, Supervision, Conceptualization. **Peter Kepplinger:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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